



MINING DEALER BEST PRACTICE

Tool and Process to Purge the 794AC Hydraulic System to Prevent Pump Damage after Draining Fluid

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

To prevent the hydraulic pump from sucking in air and damaging the hydraulic system it is necessary to purge air from the hydraulic lines whenever the fluid has been drained such as during commissioning, certain maintenance, and component changeouts.

The procedure from Caterpillar to pressurize the hydraulic tank and purge the air from the pump's hydraulic system for the 794AC is not clear and there is no single tool for efficiently isolating the breather and pressuring the system to ultimately purge the air.

2.0 Best Practice Description

The Tooling and Process described throughout this document is a Best Practice used to prevent damage/wear and shortened component life by efficiently purging air from the hydraulic system components and lines during commissioning, certain maintenance, and major component changeout.

Sotreq successfully applied this Best Practice during the commissioning of a fleet of 794AC Trucks in Northern Brazil. The customer acknowledged the success and efficiency of the process and mandated it for future hydraulic system maintenance.

Please see the instructions and reference images below for the air purging process.

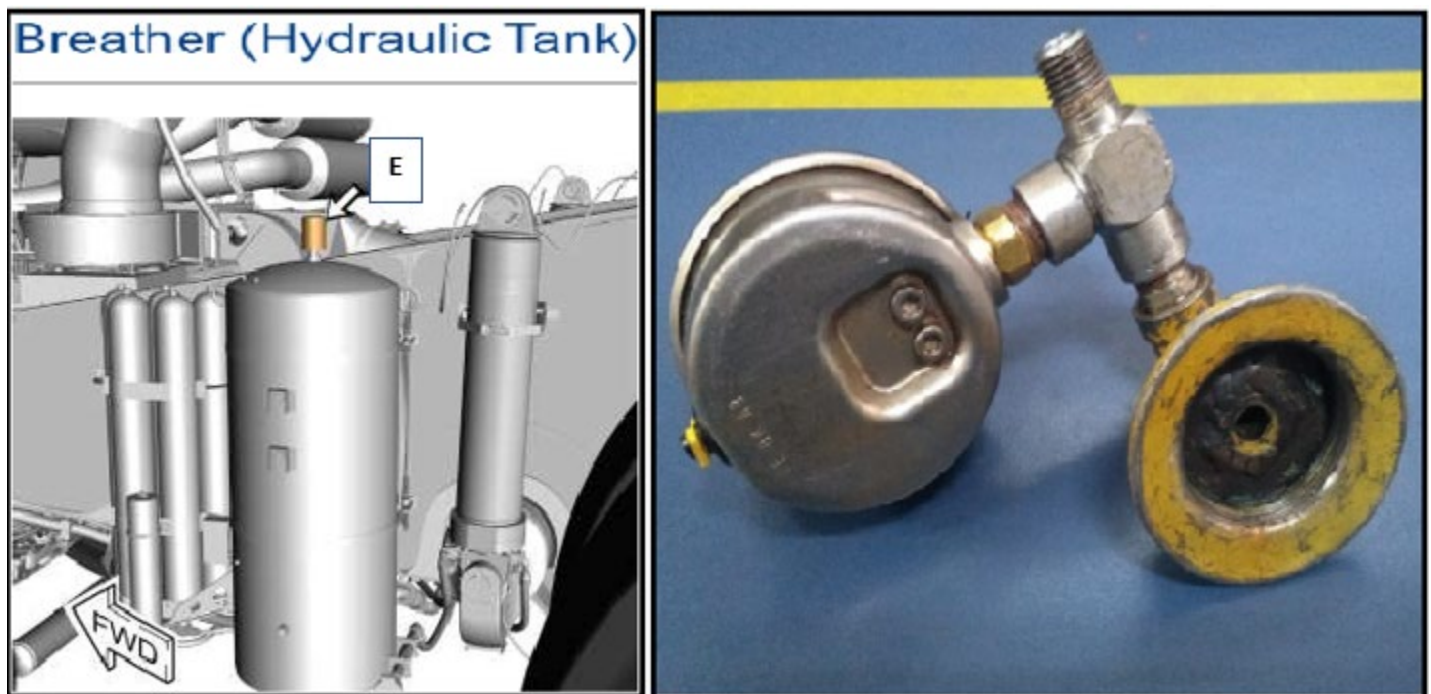


Figure 1 – View of the hydraulic tank and adapter

1. Temporarily remove the breather (E) at the top of the hydraulic tank and cover the hole. See Breather (Hoist and Brake Tank) - Replace (SEBU9099-08). If necessary, use tool 185-3630 to remove the breather.

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Figure 2 – Removing the breather

2. Insert an air adapter and regulator into the vent location.
3. Pressurize the hydraulic tank to 70 kPa (10 psi) to hydraulic priming system. Do not add pressure on the hydraulic system. (70 kPa/10 psi Maximum)

3.0 Implementation Steps

Sotreq used a filter (4T-6788) to develop the tool. The filter is cut, and part of the filter head (threaded plate) is used. Additionally, a plate with a pressure tap is installed.

With this tool ready, the breather at the top of the hydraulic tank must be removed, allowing the tool to be installed in place of the breather. The pressure entry must be connected to the pressure inlet to the air hose, with a pressure regulator limited to 10 psi.

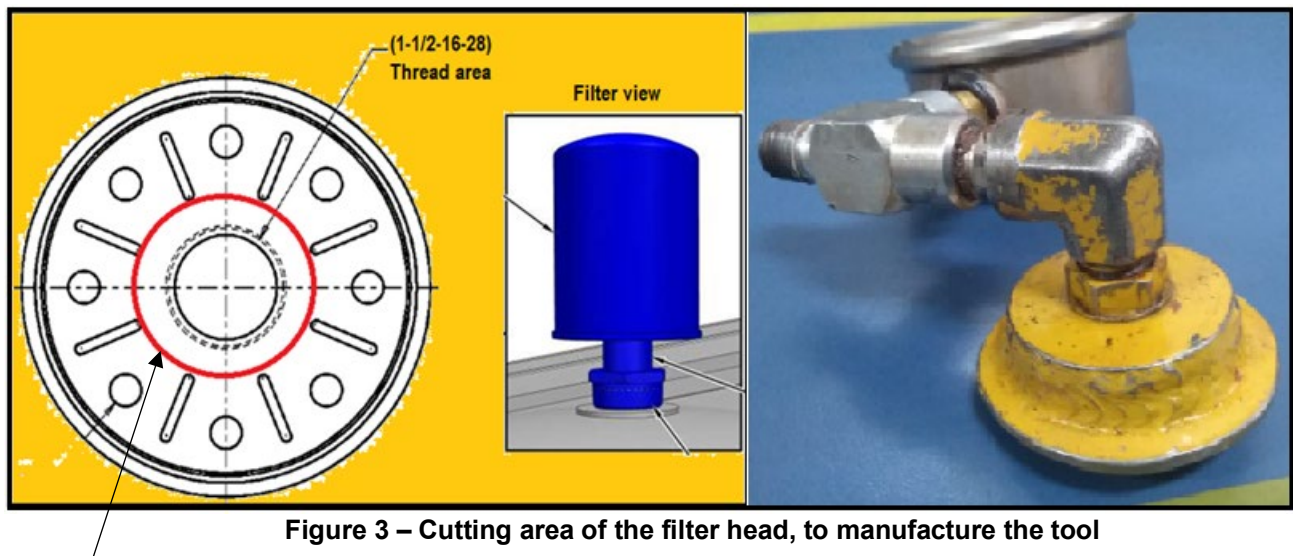


Figure 3 – Cutting area of the filter head, to manufacture the tool

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Figure 4 – Cutting the filter head

After the flange cut, weld a thick washer to the top of the flange. After that, install an elbow (085-7356) on top of the thick washer with thread for the inlet of a pressure tap with a pressure regulator (4C-8852).



Figure 5 – Flange with and without the washer and elbow adapter



Figure 6 – Manufactured tool without and with the pressure regulator

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Figure 7 – Usage of the manufactured tool during the hydraulic tank pressurization process (Max. 10 psi)

4.0 Benefits

- Performance agility for the air purging process.
- Low manufacturing cost of the tool.
- Easy storage.

5.0 Resources Required

Item	Quantity
Used Filter (it will be discarded)	1
Quick Coupling P/N 6V-3965	1
Elbow Adapter P/N 085-7356	1
Labor, Welder	1 hour

6.0 Supporting Attachments / References

N/A.

7.0 Related Best Practices

N/A.

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8.0 Acknowledgements

SOTREQ (Northern Brazil) would like to express its deep admiration for the professionals involved in improving the maintenance and repair processes developed by Mining Engineering.

SOTREQ would also like to thank the indispensable technical contribution of Doug Friedrich of Caterpillar Inc.

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